Performance Testing

Date	02 October 2026
Team ID	SWTID-2026-1529
Project Name	EduGenie: Google gemini powered learning assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	[e.g., JMeter, Locust, k6, Postman]
Type of Testing	[e.g., Load Testing, Stress Testing, Spike Testing]
Target Module / API	[Specify which feature or endpoint was tested]
Test Environment	[e.g., Local / Cloud / Hosted Server]
Test Date 02 October 2026	Google Lighthouse & K6

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	API Latency (Q&A)	150 concurrent users asking questions		< 2 seconds response time
2	Quiz Generation	Generate 5 MCQ JSONs for heavy text		JSON format validation 100%
3	Page Load	Lighthouse score on Glassmorphic UI		Performance score > 90
3				
4				

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	API Latency	< 2000 ms	1250 ms (Avg)	Pass	Gemini 3.5 Flash is fast
2	Quiz Format	100% Valid JSON	100% Valid JSON	Pass	Using strict system prompt
3	Lighthouse	> 90 Score	98 Score	Pass	Next.js 15 optimizations
1	Response Time (Avg)	< 2 seconds			
2	Response Time (Max)	< 5 seconds			
3	Throughput (Req/sec)				
4	Error Rate	< 1%			
5	CPU Utilization	< 80%			
6	Memory Utilization	< 80%			

The system performed excellently under load. FastAPI handled concurrent connections seamlessly, while the Google Gemini API provided highly accurate JSON responses within acceptable timeframes. Lighthouse testing showed a 98/100 performance score on the Glassmorphic UI.

Step 4: Observations & Analysis

Key Findings:

[Describe what the test results revealed about your system's performance]

Bottlenecks Identified:

[List any performance bottlenecks observed during testing]

Optimization Steps Taken:

[Describe any changes or optimizations made to improve performance]

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]